

# Session 14 Quiz - Natural Language Processing (NLP)

Track B | Grades 9–12 | 20 Questions

**Q1** [Type: MCQ | Difficulty: Easy | QID : 53574 ]

The branch of AI focused on understanding and generating human language is called \_\_\_\_\_.

- A. Natural Language Processing
- B. Computer Vision
- C. Generative AI
- D. Robotics

**Correct Answer: A**

**Explanation:** Natural Language Processing (NLP) is the AI subfield focused on language - it powers spell checkers, voice assistants, chatbots, and translation apps. Computer Vision handles images and video. Generative AI is a broader category that includes NLP-based generators (text) and others (images, audio). Robotics is unrelated.

**Q2** [Type: MCQ | Difficulty: Easy | QID : 53577 ]

The Break-stage step called \_\_\_\_\_ splits text into small pieces such as words, punctuation, or sub-words - for example, "I love AI!" becomes ["I", "love", "AI", "!"].

- A. Stemming
- B. Lemmatisation
- C. Tokenisation
- D. Sentiment analysis

**Correct Answer: C**

**Explanation:** Tokenisation is the first step of the Break stage - it produces the tokens that every later step (POS tagging, NER, NLU, NLG) operates on. The AI literally cannot work with raw strings - it needs tokens. Stemming and lemmatisation come later and reduce words to base forms; sentiment analysis happens at the NLU stage.

**Q3** [Type: MCQ | Difficulty: Easy | QID : 53579 ]

An NLP task that identifies whether text expresses a positive, negative, or neutral opinion is called \_\_\_\_\_ analysis.

- A. Sentiment
- B. Format
- C. Statistical
- D. Linear

**Correct Answer: A**

**Explanation:** Sentiment analysis identifies the emotional tone of text - widely used for customer reviews, brand monitoring, and customer service. Format, statistical, and linear are unrelated terms.

**Q4** [Type: MCQ | Difficulty: Easy | QID : 53581 ]

NLP systems truly understand the meaning of human language in the same way a human reader does.

- A. True
- B. False

**Correct Answer: B (False)**

**Explanation:** NLP is pattern matching, not true understanding - it finds patterns in billions of sentences and uses them to predict what makes sense. A smart-home chatbot that turns off "the living room lights" has not learned what a light is - only that "turn off" and "lights" frequently appear together in commands. The failures on sarcasm, idioms, and ambiguity reveal that what looks like understanding is really pattern recognition.

**Q5** [Type: MCQ | Difficulty: Medium | QID : 53584 ]

Stemming and lemmatisation are essentially the same operation - both reduce words to their dictionary form using a dictionary lookup.

- A. True
- B. False

**Correct Answer: B (False)**

**Explanation:** Stemming crudely chops endings off without using a dictionary (e.g., "running" → "run") and can produce non-words. Lemmatisation uses a dictionary to return the actual base form, which is how it can map "better" → "good" or "ran" → "run." The clean test is "better → good" - only lemmatisation can do that, because you cannot reach "good" by chopping endings off "better."

**Q6** [Type: MCQ | Difficulty: Medium | QID : 53585 ]

NLP systems can struggle with sarcasm, idioms, and cultural context - language that depends on shared understanding beyond the literal meaning of words.

- A. True
- B. False

**Correct Answer: A (True)**

**Explanation:** Sarcasm, idioms, slang, regional expressions, and cultural references rely on shared context the AI may not have. "Break a leg" does not mean what the words literally say. NLP improves at these over time but still struggles, particularly with rare languages, dialects, and culturally specific expressions.

**Q7** [Type: MCQ | Difficulty: Easy | QID : 53586 ]

What is the correct order of the three stages of the NLP pipeline?

- A. Generate → Understand → Break
- B. Understand → Generate → Break
- C. Break → Understand → Generate
- D. Translate → Tokenise → Output

**Correct Answer: C**

**Explanation:** Break first (tokenise, tag, identify entities), then Understand (intent, sentiment, relationships), then Generate (produce a response). Every NLP pipeline follows this order, even when it is invisible to the user.

**Q8** [Type: MCQ | Difficulty: Medium | QID : 53587 ]

An AI identifies "Satya Nadella" as a person and "Microsoft" as an organisation in a sentence. Which NLP technique is being used?

- A. Sentiment Analysis
- B. Named Entity Recognition (NER)
- C. Part-of-Speech (POS) Tagging
- D. Natural Language Generation (NLG)

**Correct Answer: B**

**Explanation:** NER is specifically designed to find and classify named entities like people, places, dates, and organisations. Sentiment analysis detects emotional tone. POS tagging labels grammatical roles (noun, verb, etc.). NLG generates new text rather than analysing existing text.

**Q9** [Type: MCQ | Difficulty: Medium | QID : 53588 ]

NLU stands for Natural Language Understanding and NLG stands for Natural Language Generation. Which task falls under NLG?

- A. Tokenising a user's question into pieces
- B. Detecting whether a sentence is positive, negative, or neutral
- C. Tagging "cat" as a noun in a sentence
- D. Producing a chatbot's text reply, one token at a time

**Correct Answer: D**

**Explanation:** NLG is the Generate stage - producing new text by predicting tokens one at a time. Tokenising belongs to Break; sentiment analysis is NLU; POS tagging is Break. Only producing the reply is NLG.

**Q10** [Type: MCQ | Difficulty: Medium | QID : 53589 ]

How does a "transformer" model, like the one used by ChatGPT, improve the AI's understanding of language?

- A. It converts all text into images so the model can "see" the patterns visually on screen
- B. It analyses each word one by one in a strict sequence to ensure no word is ever missed
- C. It uses "attention" to look at all words at once and decide what matters most
- D. It memorises every dictionary definition to avoid the need for stemming or lemmatisation steps

**Correct Answer: C**

**Explanation:** Transformers use an attention mechanism to weigh the relationships between words regardless of their distance in a sentence. Attention lets the model link "cat" near the start of a sentence to "jumped" near the end. The other options describe mechanisms transformers do not actually use.

**Q11** [Type: MCQ | Difficulty: Medium | QID : 53590 ]

You want to take a sentence you already wrote and rephrase it in two or three different styles without changing the meaning. Which tool is the best fit for this NLP task?

- A. NotebookLM
- B. Google Translate
- C. A POS tagger
- D. QuillBot

**Correct Answer: D**

**Explanation:** QuillBot is the paraphrasing tool - it rewrites your own text in modes like Standard, Fluency, or Creative while preserving meaning. NotebookLM is for source-grounded answers, Google Translate is for converting between languages, and a POS tagger only labels grammatical roles - none of those rephrase your writing while keeping meaning.

**Q12** [Type: MCQ | Difficulty: Medium | QID : 53591 ]

Which of the following is NOT a typical NLP task?

- A. Translating text from one language to another
- B. Detecting whether an email is spam
- C. Identifying the topic of a news article
- D. Calculating the area of a triangle

**Correct Answer: D**

**Explanation:** Geometric calculations from numerical inputs are math, not language processing - no text to process. Translation, spam classification, and topic identification all involve analysing or generating language, which is the domain of NLP.

**Q13** [Type: MCQ | Difficulty: Medium | QID : 53592 ]

The material emphasises that NLU is "pattern matching, not true understanding." Which example BEST illustrates this distinction?

- A. The chatbot uses bullet points instead of writing full paragraphs in its replies
- B. The chatbot turns off the lights when asked, but does not know what a light actually is
- C. The chatbot replies in English by default unless specifically asked to use another language
- D. The chatbot takes a few seconds to respond before generating the user's answer

**Correct Answer: B**

**Explanation:** The system correctly executes the request not because it conceptually understands lights, but because it matched the linguistic pattern to similar commands in its training. Pattern matching can succeed without true comprehension - and it can fail when the pattern is unusual. The other options describe surface features that have nothing to do with understanding.

**Q14** [Type: MCQ | Difficulty: Hard | QID : 53593 ]

Which of the following is a "low-resource language" challenge in NLP?

- A.** The speakers of the language refuse to use voice assistants or other AI-powered tools
- B.** The AI uses too much electricity to process complex sentences in those languages
- C.** The language has too many letters in its alphabet for the AI to tokenise properly
- D.** There is not enough training data available for certain indigenous or regional languages

**Correct Answer: D**

**Explanation:** NLP models perform poorly on languages that do not have massive amounts of digital text available for the AI to learn from. Gmail Smart Reply, Google Translate, and similar tools work well in English partly because so much English text exists online for training. Many African and Indigenous languages are dramatically underserved by current NLP systems.

**Q15** [Type: MCQ | Difficulty: Hard | QID : 53594 ]

A bank uses NLP sentiment analysis to monitor customer service calls and route "frustrated" callers to senior staff. After three months, a manager notices that escalation rates are much lower for calls from certain customer demographics despite complaint volumes being similar across groups. What is the most likely explanation?

- A.** Senior staff are selectively accepting escalations from certain demographic groups while rejecting others based on personal bias
- B.** Customers from those demographics are genuinely less frustrated and therefore not triggering the system's escalation threshold
- C.** The sentiment model was trained on data that does not reflect how frustration sounds across different accents and speech patterns
- D.** The routing system has a technical error that randomly fails to escalate calls regardless of the detected sentiment level

**Correct Answer: C**

**Explanation:** The training data limitation creates real inequity: customers whose communication style does not match what the system was trained on get misclassified more often. This pattern recurs across many NLP applications and is one of the main reasons systems trained on dominant-language text underserve users from other communities.

**Q16** [Type: MCQ | Difficulty: Hard | QID : 53595 ]

A hospital's NLP system extracts patient information from doctors' notes - medications, allergies, and conditions - with a reported 94% accuracy rate. The administrator argues this is accurate enough to use for clinical decisions without routine human review. What is the strongest argument against this proposal?

- A.** 94% accuracy across thousands of daily extractions means hundreds of errors per day, and a missed allergy is a patient, not a statistic
- B.** NLP accuracy degrades over time as medical language and terminology evolve, so 94% will drop significantly within just a few months
- C.** Doctors will lose trust in the system if they are removed from the review process and stop relying on it for clinical decisions
- D.** Regulatory bodies currently require human sign-off on all medical record extractions, making this proposal legally non-compliant

**Correct Answer: A**

**Explanation:** The concrete failure rate plus the high cost of clinical errors together justify routine human review. NLP helps with the volume; humans catch the errors that matter. The other options raise real concerns but do not target the central issue, which is that a small error rate at scale produces a large number of patient-level harms.

**Q17** [Type: MCQ Scenario | Difficulty: Medium | QID : 53596 ]

Scenario: You say to a voice assistant, "Set a timer for 5 minutes." The assistant tokenises your speech, figures out that the intent is "set timer" with duration "5 minutes," and then says aloud, "Sure - your 5-minute timer has started."

Question: Match the three pieces of work to the correct NLP stages.

- A.** Tokenising = Generate (NLG); intent detection = Break; saying the reply = Understand (NLU)
- B.** Tokenising = Break; intent detection = Understand (NLU); saying the reply = Generate (NLG)
- C.** All three pieces of work happen entirely within the Generate (NLG) stage
- D.** All three pieces of work happen entirely within the Understand (NLU) stage

**Correct Answer: B**

**Explanation:** This maps directly onto the Break → Understand → Generate pipeline. Tokenisation is the first Break step; figuring out the intent and the timer duration is NLU; producing the spoken reply is NLG. Recognising this mapping is the "looking under the hood" skill the material is building.

**Q18** [Type: MCQ Scenario | Difficulty: Hard | QID : 53597 ]

Scenario: A friend texts a chatbot: "Helping my little sister with her science fair project is a piece of cake." The chatbot replies with a recipe for cake, completely missing what the friend meant.

Question: Which NLP challenge is most clearly responsible for this failure?

**A.** Long-range dependencies

**B.** Idioms

**C.** Cultural context

**D.** Tokenisation

**Correct Answer: B**

**Explanation:** "A piece of cake" is an idiom meaning "easy" - the literal meaning has nothing to do with dessert. NLP systems that pattern-match on words can be tripped up by idioms because the meaning is not the sum of the individual words. This is the same trap as "it's raining cats and dogs."

**Q19** [Type: FRQ - Short Answer | Difficulty: Hard | QID : 53598 ]

Scenario: A social media platform deploys an NLP system to detect and remove hate speech. After deployment, the platform finds the system removes far more posts written in African American Vernacular English (AAVE) than posts written in standard English, even when the AAVE posts do not contain hate speech.

Question: Explain why this might be happening, and describe one thing the platform could do to address the problem.

*(Respond in 3–4 sentences.)*

**Model Answer:** The NLP system was likely trained primarily on standard English text and has learned to associate AAVE patterns with content the system was told to remove - either because the training data labels were biased, or because the system has not learned to distinguish between AAVE used naturally and language patterns it was trained to flag. The platform could retrain the system using more diverse language data with accurate labels for AAVE content and have human moderators with relevant cultural and linguistic background regularly audit the system's decisions to catch and correct biased classifications.

**Q20** [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53599 ]

Scenario: A regional bank is considering deploying an NLP system to automatically screen written loan applications and flag those with "risk indicators" for review. The bank's CEO says: "This will save enormous time and let us serve more customers." The compliance officer worries that NLP-based screening could be biased against certain applicant groups in ways that are hard to detect.

Question: Identify one significant benefit AND one significant risk of using NLP to screen loan applications, and describe what the bank should require to ensure the system serves applicants fairly.

*(Respond in 3–4 sentences.)*

**Model Answer:** A significant benefit is that NLP screening can quickly process large volumes of applications and flag specific concerns for human review, allowing loan officers to focus their attention on applications that need careful consideration. A significant risk is that NLP systems can learn discriminatory patterns from training data - especially if past lending decisions contained bias against certain groups - and these patterns can be difficult to detect because the system makes decisions based on language patterns rather than explicit protected characteristics. To ensure fairness, the bank should require regular auditing of decisions broken down by applicant demographics, transparent criteria for what counts as a "risk indicator," and human review of every flagged application with explicit authority to override the system - never using NLP screening alone to deny applications.